



ECEN 314 - Signals and Systems

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Module 19 - The Fourier Transform in Discrete Time

Learning Objectives



- Understand the difference between DTFT and DFT.
- Understand the relationship between the CTFT and the DFT.
- Be able to use the DFT and IDFT formulas to find Fourier transforms and inverse Fourier transforms of discrete time signals.
- Understand zero-padding and what impact it has on DFTs and inverse DFTs.

Discrete Time Fourier Transform



We have seen previously that if we sample a continuous time signal at a sufficiently high sampling rate, all of the information contained in the continuous time signal is contained in its samples.

It stands to reason that we should be able to learn about its spectrum by performing appropriate mathematical operations on its samples. We can approximate the Fourier Transform integral in terms of a Riemann sum

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt \approx \sum_{k=-\infty}^{\infty} x(k\Delta t) e^{-j2\pi f k\Delta t} \Delta t.$$

If we take Δt to be the sampling interval, T_s , and then let $x(k\Delta t) \rightarrow x[k]$ represent the discrete time samples of the signal,

$$X(f) \approx T_s \sum_{k=-\infty}^{\infty} x[k] e^{-j2\pi k f / f_s}.$$

If we work with a normalized frequency, $\tilde{f} = f/f_s$ and a normalized spectrum, $\tilde{X}(\tilde{f}) = f_s X(f)$ then

$$\tilde{X}(\tilde{f}) = \sum_{k=-\infty}^{\infty} x[k] e^{-j2\pi k \tilde{f}}.$$

This is known as the discrete-time Fourier transform (DTFT).

Discrete Time Fourier Transform

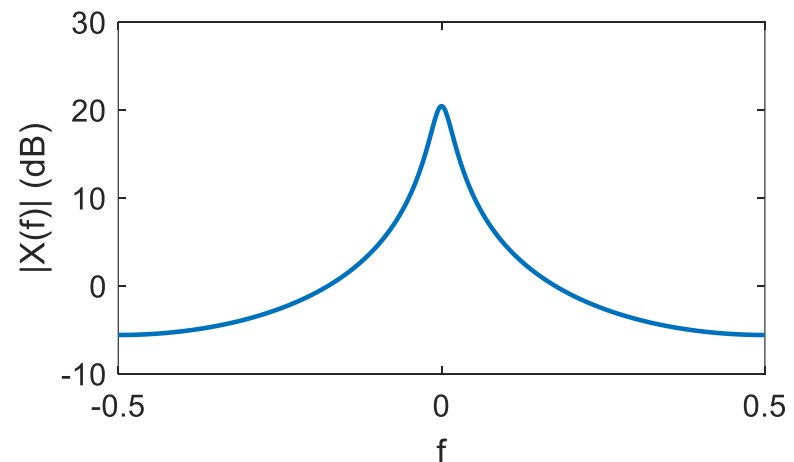
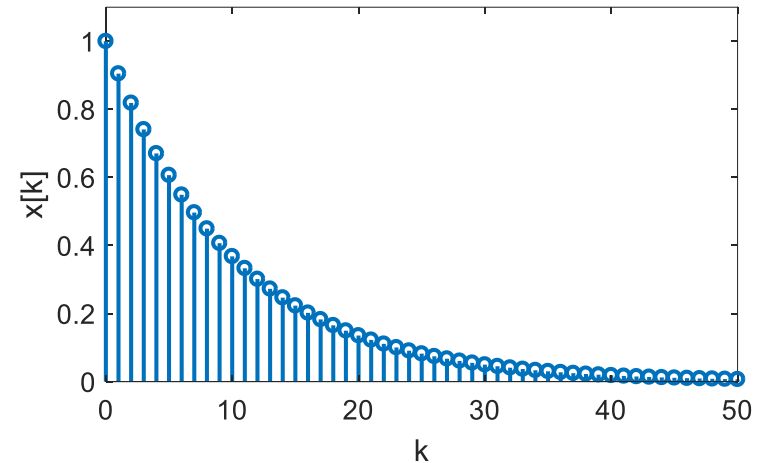


It is common to drop the tildas (in which case the normalization is implied) and write the DTFT as

$$X(f) = \sum_{k=-\infty}^{\infty} x[k] e^{-j2\pi kf}.$$

Note that:

- With the DTFT, the signal in the time domain is discrete while the signal in the frequency domain is continuous.
- Since the time domain signal was sampled at some rate of f_s , it is understood that $f_s > 2B \rightarrow B < f_s/2$. So the signal should have no frequency components beyond $|f| < f_s/2$. In terms of normalized frequency, this corresponds to $|f| < 1/2$.
- $X(f)$ is periodic in f with a period of 1.



Discrete Fourier Transform



We won't work with the DTFT much as it is more common to work with a Fourier transform of a discrete time signal which is also discrete in the frequency domain. Going back to

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt \approx \sum_{k=-\infty}^{\infty} x(k\Delta t) e^{-j2\pi f k\Delta t} \Delta t,$$

and sampling in the frequency domain as well (at a spacing of Δf),

$$X[m] = X(m\Delta f) = \sum_{k=-\infty}^{\infty} x[k] e^{-j2\pi m k \Delta t \Delta f} \Delta t.$$

It is common to choose the spacing of the frequency and time samples to be related by $\Delta t \Delta f = 1/N$ for some (usually large) integer value of N . This means that the spacing between frequency samples is $\Delta f = \frac{1}{N\Delta t} = f_s/N$.

Since the frequency content is limited to $-\frac{f_s}{2} < f < \frac{f_s}{2}$, there will then be N samples of the spectrum over the range of interest.

Replacing $\Delta t \Delta f$ with $1/N$, and normalizing by Δt results in the discrete Fourier transform (DFT)

$$X[m] = \sum_{k=-\infty}^{\infty} x[k] e^{-j2\pi m k / N}.$$

Discrete Fourier Transform



$$X[m] = \sum_{k=-\infty}^{\infty} x[k] e^{-j2\pi mk/N}.$$

- The DFT is periodic with period N since

$$\begin{aligned} X[m + N] &= \sum_{k=-\infty}^{\infty} x[k] e^{-j2\pi(m+N)k/N} = \sum_{k=-\infty}^{\infty} x[k] e^{-j2\pi mk/N} e^{-j2\pi k} \\ &= \sum_{k=-\infty}^{\infty} x[k] e^{-j2\pi mk/N} = X[m]. \end{aligned}$$

- Since $X(f)$ was non-zero only for $-\frac{f_s}{2} < f < \frac{f_s}{2}$, the spectrum of the signal is contained within N samples or one period of the DFT.
- All the information in the spectrum of the signal is contained in one period of the DFT. The rest is just a periodic extension of that spectrum.
- We can take any N consecutive samples and call it the DFT. It is common to take the samples corresponding to $m = 0, 1, 2, \dots, N - 1$, so that

$$X[m] = \sum_{k=-\infty}^{\infty} x[k] e^{-j2\pi mk/N}, \text{ for } m = 0, 1, 2, \dots, N - 1.$$

- This is referred to as the N -point DFT.

The Inverse DFT



To find the inverse DFT, take the DFT equation: $X[m] = \sum_{n=-\infty}^{\infty} x[n] e^{-j2\pi mn/N}$,

Multiply both sides by $e^{j2\pi mk/N}$ and sum over $m = 0, 1, 2, \dots, N-1$:

$$\sum_{m=0}^{N-1} X[m] e^{j2\pi mk/N} = \sum_{m=0}^{N-1} \left(\sum_{n=-\infty}^{\infty} x[n] e^{-j2\pi mn/N} \right) e^{j2\pi mk/N}$$

Exchange the order of the double sum on the RHS:

$$\sum_{m=0}^{N-1} X[m] e^{j2\pi mk/N} = \sum_{n=-\infty}^{\infty} x[n] \left(\sum_{m=0}^{N-1} e^{\frac{j2\pi m(k-n)}{N}} \right),$$

Then evaluate the inner sum using a geometric series $\sum_{m=0}^{N-1} z^m = \frac{1-z^N}{1-z}$:

$$\sum_{m=0}^{N-1} e^{j2\pi m(k-n)/N} = \sum_{m=0}^{N-1} \left(e^{j2\pi(k-n)/N} \right)^m = \frac{1 - e^{j2\pi(k-n)}}{1 - e^{j2\pi(k-n)/N}} = \begin{cases} 0, & n \neq k, \\ N, & n = k. \end{cases}$$

$$x[k] = \frac{1}{N} \sum_{m=0}^{N-1} X[m] e^{j2\pi mk/N}, \quad k = 0, 1, 2, \dots, N-1.$$

The Inverse DFT



Note that when forming the time domain samples from the N -point DFT, the resulting signal in the time domain is periodic since

$$\begin{aligned} x[k + N] &= \frac{1}{N} \sum_{m=0}^{N-1} X[m] e^{j2\pi m(k+N)/N} \\ &= \frac{1}{N} \sum_{m=0}^{N-1} X[m] e^{j2\pi mk/N} e^{j2\pi m} \\ &= \frac{1}{N} \sum_{m=0}^{N-1} X[m] e^{j2\pi mk/N} = x[k] \end{aligned}$$

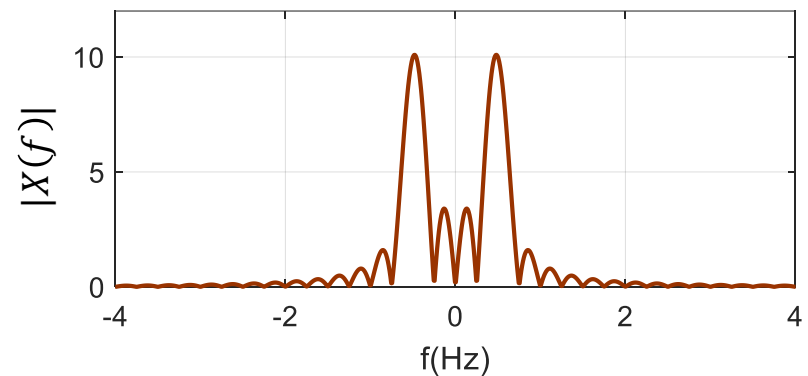
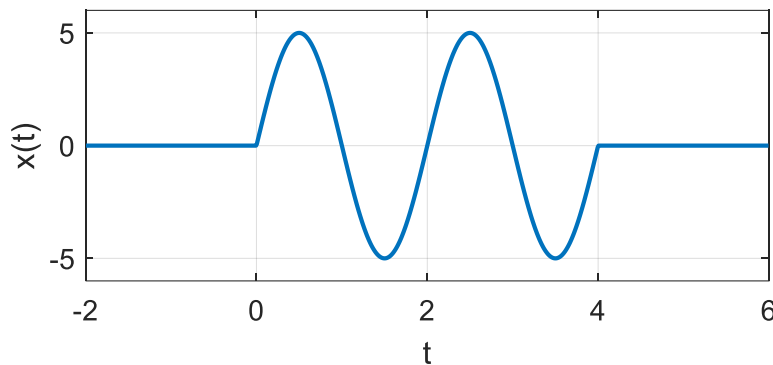
Like the DFT, the IDFT will have N distinct values. So, when working with N -point DFTs/IDFTs, it is generally assumed that the time domain signal has no more than N samples.

- ✓ If the original $x[k]$ had fewer than N samples, it is padded with zeros to form a signal with N samples.
- ✓ If the original $x[k]$ had more than N samples, it is truncated to form a signal with N samples.

Example



As an example, suppose we had a continuous time signal $x(t) = 5 \sin(\pi t) \text{rect}\left(\frac{t-2}{4}\right)$ whose CTFT is $X(f) = 10je^{-j4\pi f} [\text{sinc}(4f - 2) - \text{sinc}(4f + 2)]$.



- ✓ Strictly speaking, this signal has infinite BW, so no matter what rate we sample at there will be some aliasing.
- ✓ Practically, there is not much frequency content beyond about 2Hz, so as long as we sample at a rate of at least 4Hz, the aliasing should be minimal.
- ✓ Since the signal lasts 4 seconds in the time domain, at a sampling rate of 4Hz, we would need at least $N=16$ samples.

Example ($N=16$ samples @ 4Hz)

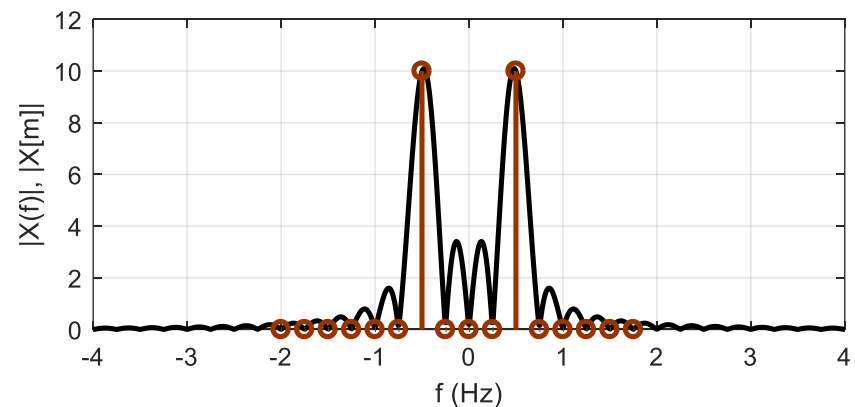
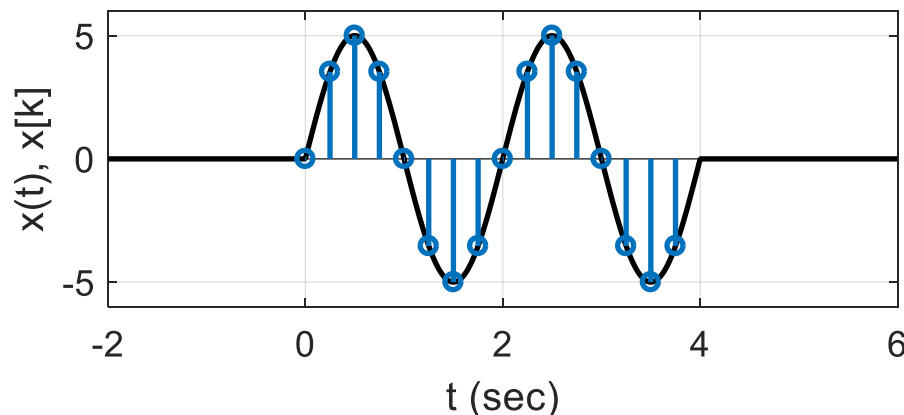


$$x[k] = x(k/4), \quad k = 0, 1, 2, \dots, 15,$$

$$\mathbf{x} = \left[0, \frac{5}{\sqrt{2}}, 5, \frac{5}{\sqrt{2}}, 0, -\frac{5}{\sqrt{2}}, -5, -\frac{5}{\sqrt{2}}, 0, \frac{5}{\sqrt{2}}, 5, \frac{5}{\sqrt{2}}, 0, -\frac{5}{\sqrt{2}}, -5, -\frac{5}{\sqrt{2}} \right]$$

$$X[m] = \sum_{k=0}^{15} x[k] e^{-j\pi mk/8}, \quad m = 0, 1, 2, \dots, 15,$$

$$\mathbf{X} = [0, 0, -40j, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 40j, 0]$$



In order to get the frequency domain pictures to line up, a few modifications were made:

- ✓ Samples corresponding to $m = -8, -7, -6, \dots, 6, 7$ were used,
- ✓ $\mathbf{X}$ was scaled by Δt ($=1/4$) as discussed in Slide 5.

Example ($N=32$ samples @ 4Hz)

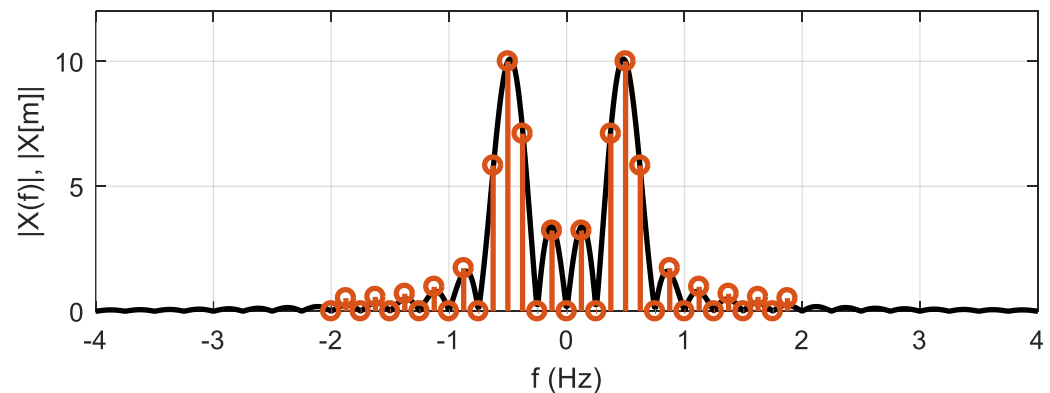
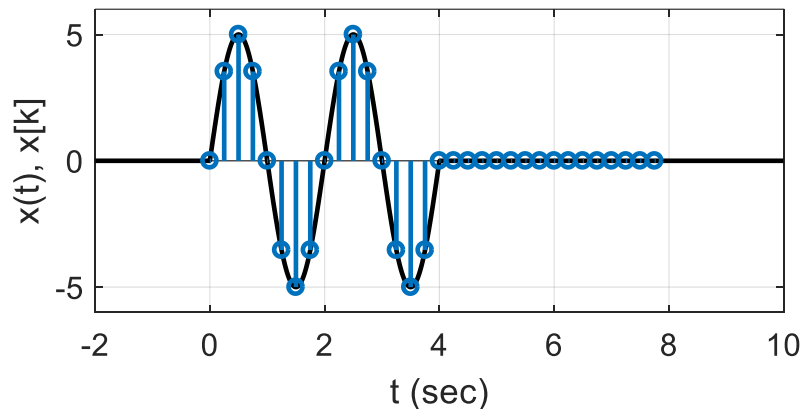


$$x[k] = x(k/4), \quad k = 0, 1, 2, \dots, 31,$$

$$\mathbf{x} = \left[0, \frac{5}{\sqrt{2}}, 5, \frac{5}{\sqrt{2}}, 0, -\frac{5}{\sqrt{2}}, -5, -\frac{5}{\sqrt{2}}, 0, \frac{5}{\sqrt{2}}, 5, \frac{5}{\sqrt{2}}, 0, -\frac{5}{\sqrt{2}}, -5, -\frac{5}{\sqrt{2}}, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \right]$$

$$X[m] = \sum_{k=0}^{31} x[k] e^{-j\pi m k / 16}, \quad m = 0, 1, 2, \dots, 31,$$

$$\mathbf{X} = \begin{bmatrix} 0, -2.0946, 0, -2.2979, 0, -2.8, 0, -3.9188, 0, -6.9051, 0, -23.3312, \\ 40j, 28.4292, 0, 12.9186, 0, 12.9186, 0, 28.4292, -40j, \\ -23.3312, -6.9051, 0, -3.9188, 0, -2.8, 0, -2.2979, 0, -2.0946 \end{bmatrix}$$



By zero-padding the signal in the time domain, we get a higher resolution of samples in the frequency domain.

Example ($N=32$ samples @ 8Hz)

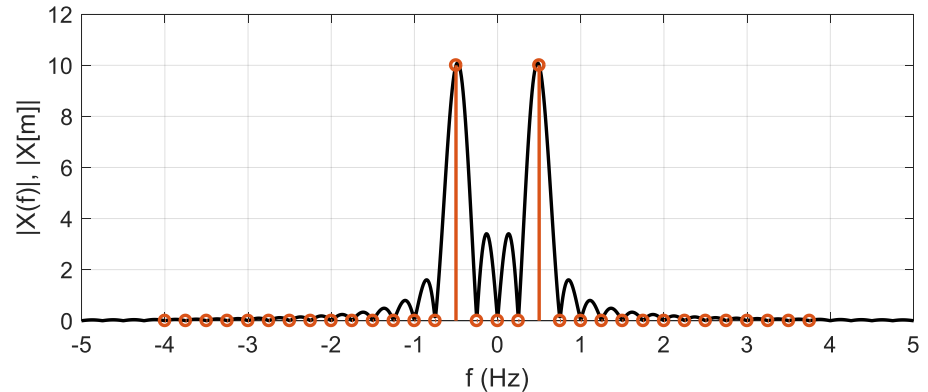
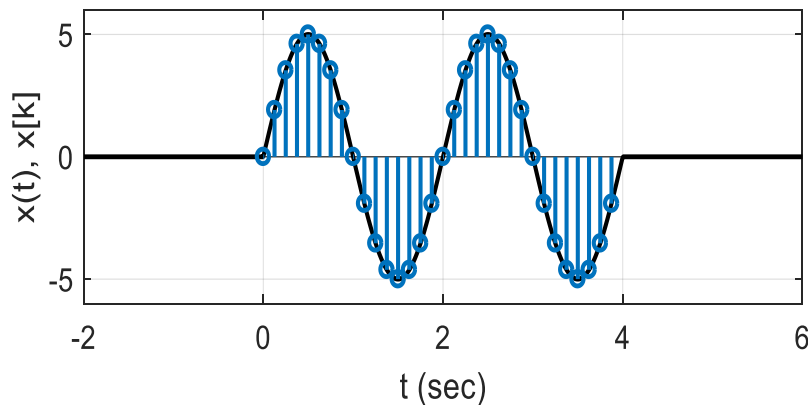


$$x[k] = x(k/4), \quad k = 0, 1, 2, \dots, 31,$$

$$\mathbf{x} = \left[0, \frac{5}{\sqrt{2}}, 5, \frac{5}{\sqrt{2}}, 0, -\frac{5}{\sqrt{2}}, -5, -\frac{5}{\sqrt{2}}, 0, \frac{5}{\sqrt{2}}, 5, \frac{5}{\sqrt{2}}, 0, -\frac{5}{\sqrt{2}}, -5, -\frac{5}{\sqrt{2}}, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 \right]$$

$$X[m] = \sum_{k=0}^{31} x[k] e^{-j\pi m k / 16}, \quad m = 0, 1, 2, \dots, 31,$$

$$\mathbf{X} = \begin{bmatrix} 0, -2.0946, 0, -2.2979, 0, -2.8, 0, -3.9188, 0, -6.9051, 0, -23.3312, \\ 40j, 28.4292, 0, 12.9186, 0, 12.9186, 0, 28.4292, -40j, \\ -23.3312, -6.9051, 0, -3.9188, 0, -2.8, 0, -2.2979, 0, -2.0946 \end{bmatrix}$$

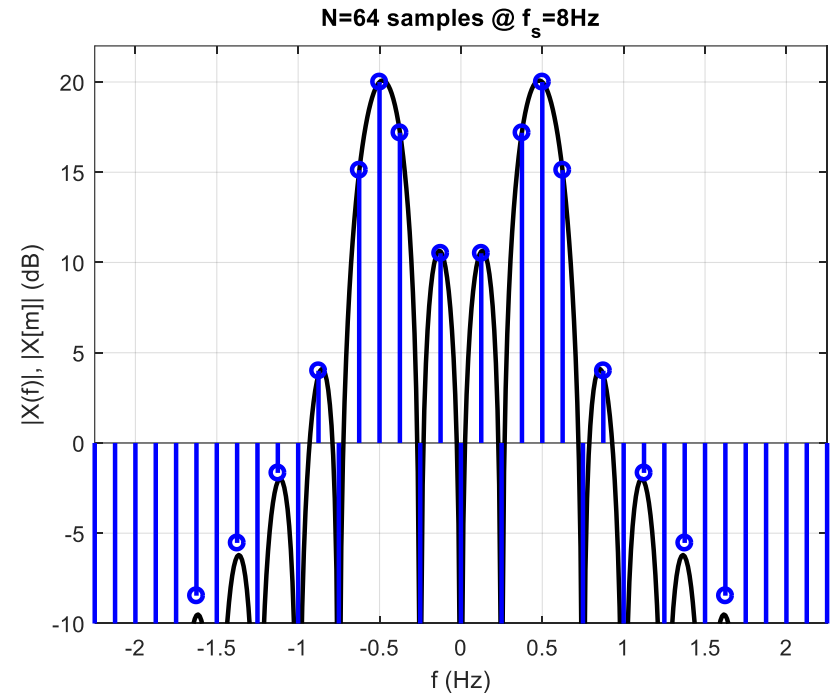
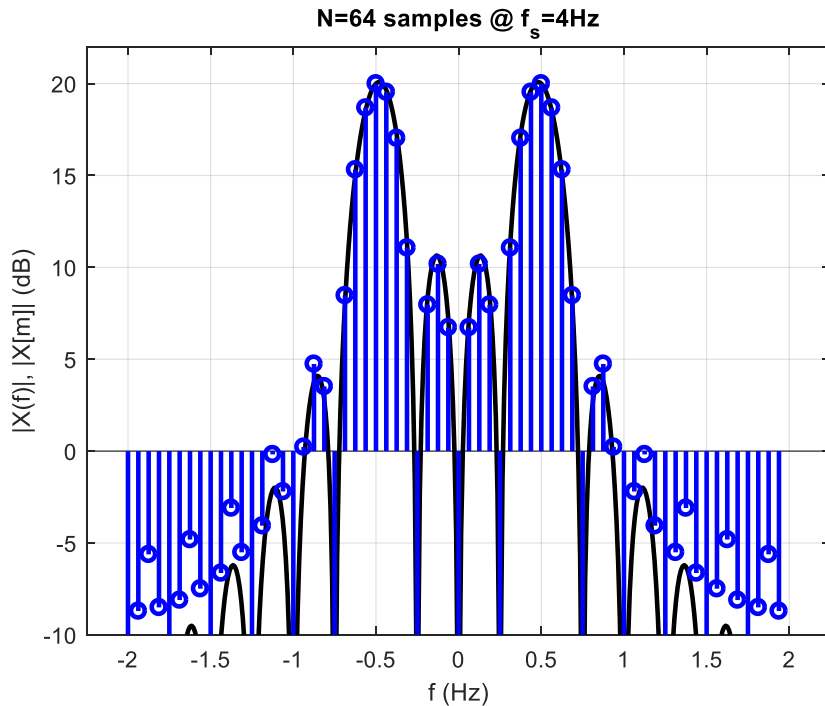


Here we got higher resolution of the signal in the time domain and what essentially acts like zero padding in the frequency domain.

Aliasing



You can see the aliasing better if you view the spectral plots on a dB scale.



Note that at the higher sampling rate, the aliasing is reduced significantly.

A Few Examples of DFTs



Example 19.1: Suppose $x(t) = e^{-at}u(t)$ (one-sided exponential)

Taking N samples with spacing Δt produces the discrete time signal

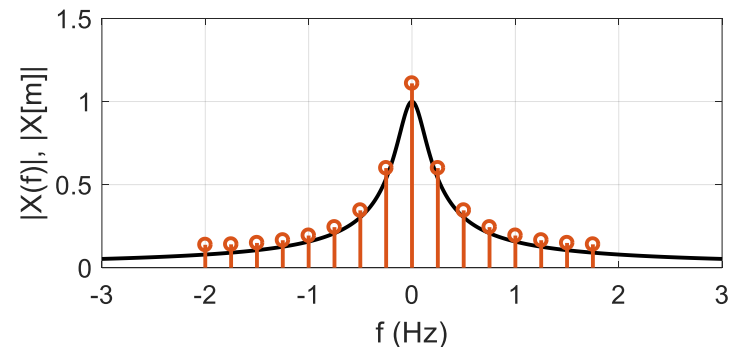
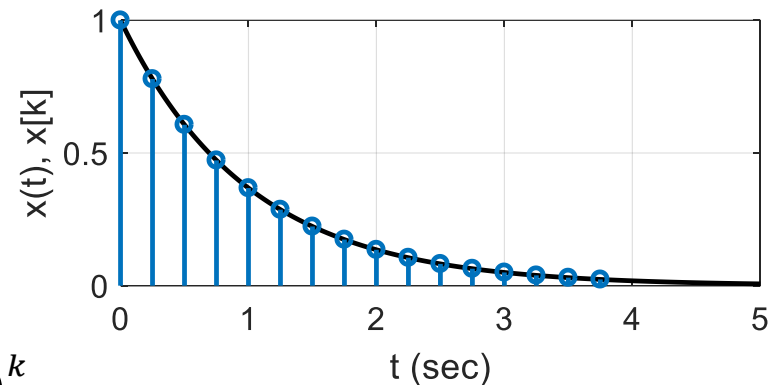
$$x[k] = e^{-a\Delta tk}, \quad k = 0, 1, 2, \dots, N-1.$$

The DFT is then:

$$X[m] = \sum_{k=0}^{N-1} e^{-a\Delta tk} e^{-j2\pi km/N} = \sum_{k=0}^{N-1} \left(e^{-(a\Delta t + j2\pi m/N)} \right)^k$$

This series can be simplified using the geometric series identity, $\sum_{k=0}^{N-1} z^k = \frac{1-z^N}{1-z}$ so that

$$\begin{aligned} X[m] &= \frac{1 - \left(e^{-(a\Delta t + j2\pi m/N)} \right)^N}{1 - e^{-(a\Delta t + j2\pi m/N)}} \\ &= \frac{1 - e^{-a\Delta t N}}{1 - e^{-(a\Delta t + j2\pi m/N)}} \end{aligned}$$



A Few Examples of DFTs



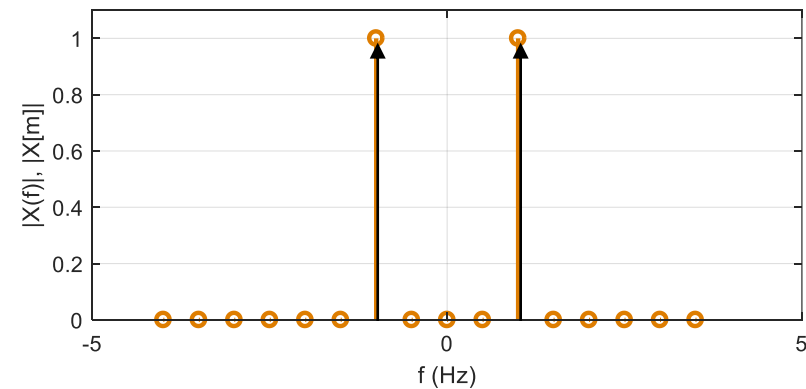
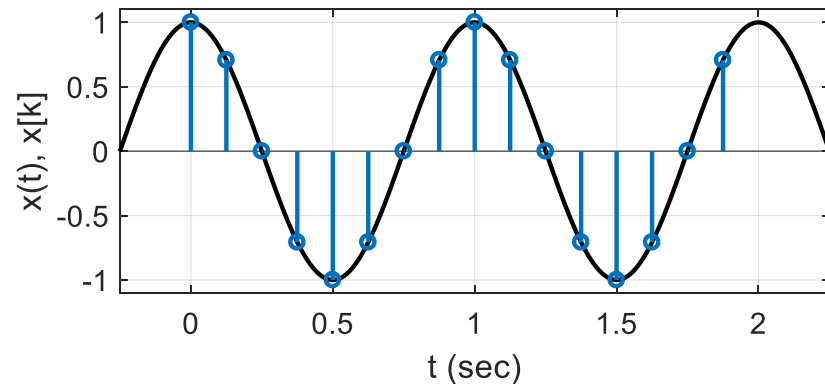
Example 19.2: Suppose $x(t) = \cos(2\pi t)$
(sinusoid)

Taking N samples with spacing Δt produces the discrete time signal

$$x[k] = \cos(2\pi k\Delta t), \quad k = 0, 1, 2, \dots, N-1.$$

The DFT is then:

$$\begin{aligned} X[m] &= \sum_{k=0}^{N-1} \cos(2\pi k\Delta t) e^{-j2\pi km/N} \\ &= \frac{1}{2} \sum_{k=0}^{N-1} e^{j2\pi k(\Delta t - \frac{m}{N})} + \frac{1}{2} \sum_{k=0}^{N-1} e^{-j2\pi k(\Delta t + \frac{m}{N})} \\ &= \frac{1 - (e^{j2\pi(\Delta t - \frac{m}{N})})^N}{1 - e^{j2\pi(\Delta t - \frac{m}{N})}} + \frac{1 - (e^{-j2\pi(\Delta t + \frac{m}{N})})^N}{1 - e^{-j2\pi(\Delta t + \frac{m}{N})}} \\ &= \frac{1 - e^{-j2\pi(m - N\Delta t)}}{1 - e^{-j2\pi(m - N\Delta t)/N}} + \frac{1 - e^{-j2\pi(m + N\Delta t)}}{1 - e^{-j2\pi(m + N\Delta t)/N}} \end{aligned}$$



$$= \frac{N}{2} \delta[m - N\Delta t] + \frac{N}{2} \delta[m + N\Delta t]$$

A Few Examples of DFTs

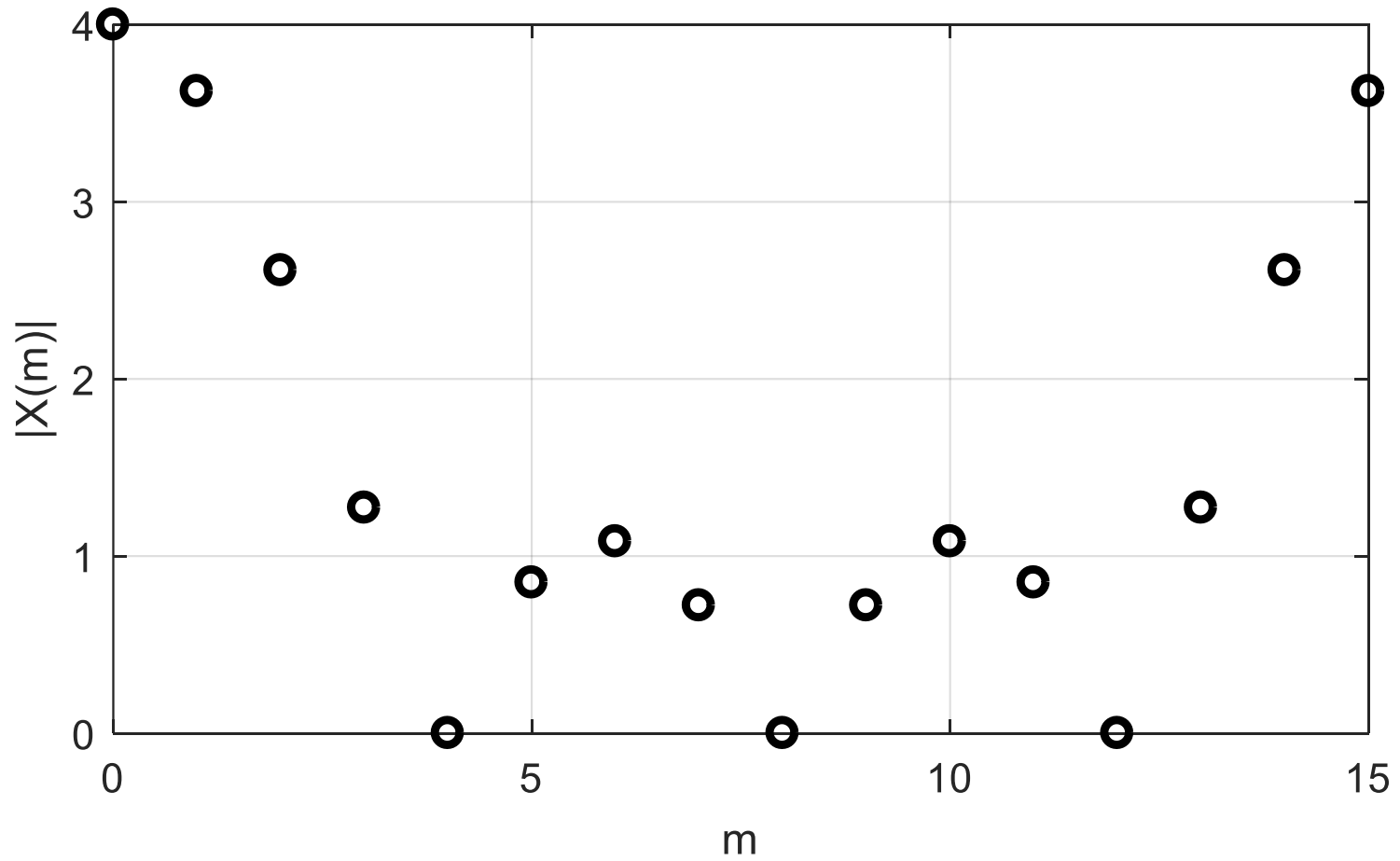


Example 19.3: Suppose $x(t) = \text{rect}(t - 1/2)$
(square pulse)

Take $N = 16$ samples at a sampling rate of
 $f_s = 4\text{Hz}$ to produce the discrete time signal

$$x = [1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0].$$

Find the DFT. Sketch the result.



Properties of DFTs



DFTs have properties that are very similar to those for the continuous time FT.

Property	Signal, $x[k]$	Transform, $X[m]$
Linearity	$ax_1[k] + bx_2[k]$	$aX_1[m] + bX_2[m]$
Time shifting	$x[k - n]$	$X[m]e^{-j2\pi mn/N}$
Frequency shifting	$x[k]e^{j2\pi kn/N}$	$X[(m - n) \bmod N]$
Time/Frequency Reversal	$x[(N - k) \bmod N]$	$X[N - m]$
Convolution	$\sum_{n=0}^{N-1} x[n]y[(k - n) \bmod N]$	$X[m] \cdot Y[m]$
Multiplication	$x[k] \cdot y[k]$	$\sum_{n=0}^{N-1} X[n]Y[(m - n) \bmod N]$
Parseval's Theorem	$\sum_{k=0}^{N-1} x[k] ^2$	$\frac{1}{N} \sum_{m=0}^{N-1} X[m] ^2$

Note: In the above table, the notation $n \bmod N$ refers to the remainder of dividing n by N and results in an integer in the range $[0, N-1]$.